Citi Bike departures, three hours ahead

A working Manhattan-total forecast using public trips and forecast weather in Databricks Free Edition.

Plan for the next three hours

Estimate served departures across Manhattan.
Publish before the first target hour.
Compare weather with a calendar fallback.

HOUR 1

T to T+1

HOUR 2

T+1 to T+2

HOUR 3

T+2 to T+3

Three separate hourly totals at each origin T

IMPLEMENTED

Hourly forecasts + dashboard + daily health
Daily releases, outcomes and station evidence

WHY START WITH MANHATTAN?

Station archive coverage depends on activity.
Reliable operation history is still needed.

From public data to a persisted forecast

Bulk trips stay on the RAID. Compact features, models, forecasts and health reports live in Databricks.

BATCH DEVELOPMENT

Public trips + archived weather

Geographic labels; released trip profiles

↓

Chronological model comparison

Initial selection + separate monthly release audit

↓

Frozen model + MLflow report

Weather Poisson + calendar fallback

HOURLY SERVING / :55 UTC

Collect live public inputs

Forecast weather; GBFS; MTA alerts archived

↓

Score the next three hours

Weather when usable; calendar otherwise

↓

Persist -> dashboard + daily health

Preserve first prediction; evaluate released outcomes

Input cutoff T-5 min | Trip release +45 days* | Weather run +7 hours* | First publication wins

*Availability assumptions; early archived weather includes hindcasts. GBFS/MTA collection is not used by the current borough model.

A useful benchmark, with clear limits

June-August 2026 release-aware backtests: 6,552 common forecast rows across three horizons.

MEAN ABSOLUTE ERROR / departures per hour

Hour-of-week profile

1,386

Monthly selected weather model

768

44.6% lower MAE than the weekly profile

Monthly selected model MAE

2026-06	**734**
2026-07	**840**
2026-08	**727**

June calendar model was better: 650 vs. 734.
Weather does not improve every month.

REMAINING FOR THE ORIGINAL PROPOSAL

Verified station history + historical alert evidence
Class deployment + documented shared catalog
Published live outcomes for production accuracy

Retrospective evidence with release/hindcast assumptions. Active model frozen; live accuracy pending.

Open live demo